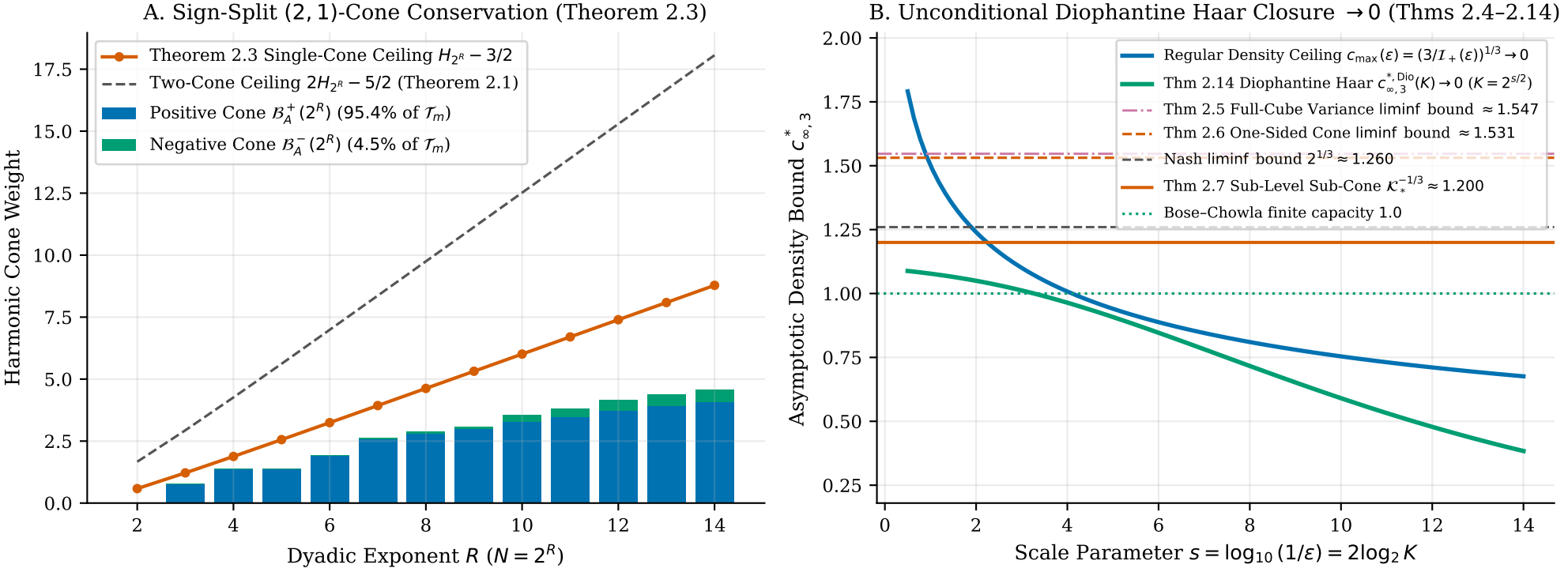
A. Sign-Split (2, 1)-Cone Conservation (Theorem 2.3)
Theorem 2.3 Single-Cone Ceiling $H_{2^R} - 3/2$
Two-Cone Ceiling $2H_{2^R} - 5/2$ (Theorem 2.1)
Positive Cone $\mathcal{B}_A^+(2^R)$ (95.4% of $\mathcal{T}_m$)
Negative Cone $\mathcal{B}_A^-(2^R)$ (4.5% of $\mathcal{T}_m$)
Harmonic Cone Weight
Dyadic Exponent R ($N = 2^R$)
B. Unconditional Diophantine Haar Closure $\to 0$ (Thms 2.4–2.14)
Regular Density Ceiling $c_{\max}(\varepsilon) = (3/\mathcal{I}_+(\varepsilon))^{1/3} \to 0$
Thm 2.14 Diophantine Haar $c_{\infty,3}^{*,\mathrm{Dio}}(K) \to 0$ ($K = 2^{s/2}$)
Thm 2.5 Full-Cube Variance liminf bound ≈ 1.547
Thm 2.6 One-Sided Cone liminf bound ≈ 1.531
Nash liminf bound $2^{1/3} \approx 1.260$
Thm 2.7 Sub-Level Sub-Cone $\mathcal{K}_*^{-1/3} \approx 1.200$
Bose–Chowla finite capacity 1.0
Asymptotic Density Bound $c_{\infty,3}^*$
Scale Parameter $s = \log_{10}(1/\varepsilon) = 2\log_2 K$